

Manual Change History (4021389) B3.9, B4.5, B5.9 Industrial Operation and Maintenance Manual

Date	Not Available
Section	Front
Procedure Number	4021389-toc (/qs3/pubsys2/xml/en/manual/4021389/4021389-toc.html)
Procedure Title	Table of Contents
Update Reason	

Date	12-Mar-2002
Section	Section i - Introduction
Procedure Number	99-204-004 (/qs3/pubsys2/xml/en/procedures/99/99-204-004.html)
Procedure Title	Symbols
Update Reason	

Date	12-Mar-2002
Section	Section i - Introduction
Procedure Number	99-204-005 (/qs3/pubsys2/xml/en/procedures/99/99-204-005.html)
Procedure Title	Illustrations
Update Reason	

Date	14-Mar-2002
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-103-999 (/qs3/pubsys2/xml/en/procedures/99/99-103-999.html)
Procedure Title	Daily Maintenance Procedures - Overview
Update Reason	

Date	14-Mar-2002
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-010-058 (/qs3/pubsys2/xml/en/procedures/99/99-010-058.html)
Procedure Title	Air Intake Piping
Update Reason	
Date	14-Mar-2002
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-012-021 (/qs3/pubsys2/xml/en/procedures/99/99-012-021.html)
Procedure Title	Air Tanks and Reservoirs
Update Reason	
Date	14-Mar-2002
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-003-018 (/qs3/pubsys2/xml/en/procedures/99/99-003-018.html)
Procedure Title	Crankcase Breather Tube
Update Reason	
Date	14-Mar-2002
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-006-043 (/qs3/pubsys2/xml/en/procedures/99/99-006-043.html)
Procedure Title	Fuel-Water Separator
Update Reason	

Date	14-Mar-2002
Section	Section 4 - Maintenance Procedures at 250 Hours or 3 Months
Procedure Number	99-999-999 (/qs3/pubsys2/xml/en/procedures/99/99-999-999.html)
Procedure Title	Maintenance Procedures - Overview
Update Reason	
Date	14-Mar-2002
Section	Section 5 - Maintenance Procedures at 500 Hours or 6 Months
Procedure Number	99-999-999 (/qs3/pubsys2/xml/en/procedures/99/99-999-999.html)
Procedure Title	Maintenance Procedures - Overview
Update Reason	
Date	14-Mar-2002
Section	Section 6 - Maintenance Procedures at 1000 Hours or 1 Year
Procedure Number	99-999-999 (/qs3/pubsys2/xml/en/procedures/99/99-999-999.html)
Procedure Title	Maintenance Procedures - Overview
Update Reason	
Date	14-Mar-2002
Section	Section 7 - Maintenance Procedures at 2000 Hours or 2 Years
Procedure Number	99-999-999 (/qs3/pubsys2/xml/en/procedures/99/99-999-999.html)
Procedure Title	Maintenance Procedure - Overview
Update Reason	

Date	14-Mar-2002
Section	Section 7 - Maintenance Procedures at 2000 Hours or 2 Years
Procedure Number	99-012-015 (/qs3/pubsys2/xml/en/procedures/99/99-012-015.html)
Procedure Title	Air Compressor Discharge Lines
Update Reason	
Date	03-Apr-2002
Section	Section TS - Troubleshooting Symptoms
Procedure Number	99-t00-002 (/qs3/pubsys2/xml/en/procedures/99/99-t00-002.html)
Procedure Title	Troubleshooting Symptoms Charts
Update Reason	
Date	08-Oct-2002
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-008-040 (/qs3/pubsys2/xml/en/procedures/99/99-008-040.html)
Procedure Title	Fan, Cooling
Update Reason	
Date	18-Feb-2003
Section	Section 5 - Maintenance Procedures at 500 Hours or 6 Months
Procedure Number	99-008-115 (/qs3/pubsys2/xml/en/procedures/99/99-008-115.html)
Procedure Title	Engine Coolant Antifreeze
Update Reason	

Date	10-Apr-2003
Section	Section 4 - Maintenance Procedures at 250 Hours or 3 Months
Procedure Number	100-005-016 (/qs3/pubsys2/xml/en/procedures/100/100-005-016.html)
Procedure Title	Fuel Pump
Update Reason	
Date	10-Apr-2003
Section	Section 6 - Maintenance Procedures at 1000 Hours or 1 Year
Procedure Number	100-008-036 (/qs3/pubsys2/xml/en/procedures/100/100-008-036.html)
Procedure Title	Fan Hub, Belt Driven
Update Reason	
Date	10-Apr-2003
Section	Section V - Maintenance Specifications
Procedure Number	100-018-008 (/qs3/pubsys2/xml/en/procedures/100/100-018-008.html)
Procedure Title	Sealants
Update Reason	
Date	16-Apr-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t081-006-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t081-006-om.html)
Procedure Title	Exhaust Smoke Excessive Under Load
Update Reason	
Date	16-Apr-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t087-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t087-om.html)
Procedure Title	Fuel Consumption Excessive
Update Reason	

Date	16-Apr-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t091-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t091-om.html)
Procedure Title	Fuel in Coolant
Update Reason	

Date	16-Apr-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t092-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t092-om.html)
Procedure Title	Fuel in the Lubricating Oil
Update Reason	

Date	16-Apr-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t093-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t093-om.html)
Procedure Title	Fuel or Lubricating Oil Leaking From Exhaust Manifold
Update Reason	

Date	16-Apr-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t096-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t096-om.html)
Procedure Title	Intake Manifold Air Temperature Above Specification
Update Reason	

Date	16-Apr-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t097-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t097-om.html)
Procedure Title	Intake Manifold Pressure (Boost) is Below Normal
Update Reason	
Date	16-Apr-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t102-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t102-om.html)
Procedure Title	Lubricating Oil Consumption Excessive
Update Reason	
Date	21-Apr-2003
Section	Section D - System Diagrams
Procedure Number	00-200-999 (/qs3/pubsys2/xml/en/procedures/00/00-200-999.html)
Procedure Title	System Diagrams - Overview
Update Reason	
Date	14-May-2003
Section	Section W - Warranty
Procedure Number	ind-qsb-qsc-q9-b-c-a (/qs3/pubsys2/xml/en/warrantyind/ind-qsb-qsc-q9-b-c-a.html)
Procedure Title	Coverage
Update Reason	
Date	13-Aug-2003
Section	Section 4 - Maintenance Procedures at 250 Hours or 3 Months
Procedure Number	40-003-024 (/qs3/pubsys2/xml/en/procedures/40/40-003-024.html)
Procedure Title	Closed Crankcase Ventilation Hoses
Update Reason	

Date	02-Sep-2003
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	100-008-039 (/qs3/pubsys2/xml/en/procedures/100/100-008-039.html)
Procedure Title	Fan Spacer and Pulley
Update Reason	
Date	09-Oct-2003
Section	Section W - Warranty
Procedure Number	ind-frt (/qs3/pubsys2/xml/en/warrantyind/ind-frt.html)
Procedure Title	California Emission Control System Warranty, Off-Highway
Update Reason	
Date	14-Oct-2003
Section	Section W - Warranty
Procedure Number	ind-bck (/qs3/pubsys2/xml/en/warrantyind/ind-bck.html)
Procedure Title	Owners Responsibility
Update Reason	
Date	25-Feb-2004
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	40-013-001-om-rep (/qs3/pubsys2/xml/en/procedures/40/40-013-001-om-rep.html)
Procedure Title	Alternator
Update Reason	
Date	03-Mar-2004
Section	Section V - Maintenance Specifications
Procedure Number	40-018-024 (/qs3/pubsys2/xml/en/procedures/40/40-018-024.html)
Procedure Title	Cummins®/Fleetguard® Filter Specifications
Update Reason	

Date	03-Mar-2004
Section	Section V - Maintenance Specifications
Procedure Number	99-018-010 (/qs3/pubsys2/xml/en/procedures/99/99-018-010.html)
Procedure Title	Fraction, Decimal, Millimeter Conversions
Update Reason	
Date	03-Mar-2004
Section	Section V - Maintenance Specifications
Procedure Number	99-018-011 (/qs3/pubsys2/xml/en/procedures/99/99-018-011.html)
Procedure Title	Newton-Meter to Foot-Pound Conversions
Update Reason	
Date	03-Mar-2004
Section	Section V - Maintenance Specifications
Procedure Number	99-018-012 (/qs3/pubsys2/xml/en/procedures/99/99-018-012.html)
Procedure Title	Pipe Plug Torque Values
Update Reason	
Date	03-Mar-2004
Section	Section V - Maintenance Specifications
Procedure Number	99-018-013 (/qs3/pubsys2/xml/en/procedures/99/99-018-013.html)
Procedure Title	Tap-Drill Chart - U.S. Customary and Metric
Update Reason	
Date	03-Mar-2004
Section	Section V - Maintenance Specifications
Procedure Number	99-018-014 (/qs3/pubsys2/xml/en/procedures/99/99-018-014.html)
Procedure Title	Weights and Measures - Conversion Factors
Update Reason	

Date	13-Jun-2005
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	100-013-020 (/qs3/pubsys2/xml/en/procedures/100/100-013-020.html)
Procedure Title	Starting Motor
Update Reason	
Date	30-Jun-2005
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	100-010-027 (/qs3/pubsys2/xml/en/procedures/100/100-010-027.html)
Procedure Title	Charge-Air Cooler
Update Reason	
Date	12-Aug-2005
Section	Section 5 - Maintenance Procedures at 500 Hours or 6 Months
Procedure Number	40-006-014 (/qs3/pubsys2/xml/en/procedures/40/40-006-014.html)
Procedure Title	Fuel Filter (Canister Type)
Update Reason	
Date	06-Feb-2006
Section	Section D - System Diagrams
Procedure Number	100-200-006 (/qs3/pubsys2/xml/en/procedures/100/100-200-006.html)
Procedure Title	Flow Diagram, Compressed Air System
Update Reason	

Date	15-Feb-2006
Section	Front
Procedure Number	refno (/qs3/pubsys2/xml/en/procedures//refno.html)
Procedure Title	Important Reference Numbers
Update Reason	
Date	09-Mar-2006
Section	Section 1 - Operating Instructions
Procedure Number	40-101-018 (/qs3/pubsys2/xml/en/procedures/40/40-101-018.html)
Procedure Title	Starting Procedure After Extended Shutdown or Oil Change
Update Reason	
Date	09-Mar-2006
Section	Section 5 - Maintenance Procedures at 500 Hours or 6 Months
Procedure Number	40-007-002 (/qs3/pubsys2/xml/en/procedures/40/40-007-002.html)
Procedure Title	Lubricating Oil and Filters
Update Reason	
Date	31-Mar-2006
Section	Section 4 - Maintenance Procedures at 250 Hours or 3 Months
Procedure Number	40-008-047 (/qs3/pubsys2/xml/en/procedures/40/40-008-047.html)
Procedure Title	Radiator Pressure Cap
Update Reason	

Date	31-Mar-2006
Section	Section 7 - Maintenance Procedures at 2000 Hours or 2 Years
Procedure Number	40-008-018-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-008-018-om-ind.html)
Procedure Title	Cooling System
Update Reason	
Date	31-Mar-2006
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	40-006-003-om-rep (/qs3/pubsys2/xml/en/procedures/40/40-006-003-om-rep.html)
Procedure Title	Air in Fuel
Update Reason	
Date	24-Apr-2006
Section	Section 6 - Maintenance Procedures at 1000 Hours or 1 Year
Procedure Number	40-003-004-om (/qs3/pubsys2/xml/en/procedures/40/40-003-004-om.html)
Procedure Title	Overhead Set
Update Reason	
Date	04-May-2006
Section	Section i - Introduction
Procedure Number	99-204-001 (/qs3/pubsys2/xml/en/procedures/99/99-204-001.html)
Procedure Title	To the Owner and Operator
Update Reason	

Date	22-May-2006
Section	Front
Procedure Number	om-frwd (/qs3/pubsys2/xml/en/manual/preface/om-frwd.html)
Procedure Title	Foreword
Update Reason	
Date	31-May-2006
Section	Section S - Service Assistance
Procedure Number	99-202-005 (/qs3/pubsys2/xml/en/procedures/99/99-202-005.html)
Procedure Title	Division and Regional Offices
Update Reason	
Date	27-Jun-2006
Section	Section L - Service Literature
Procedure Number	99-205-002 (/qs3/pubsys2/xml/en/procedures/99/99-205-002.html)
Procedure Title	Service Literature Ordering Location
Update Reason	
Date	24-Jul-2006
Section	Section S - Service Assistance
Procedure Number	99-202-008 (/qs3/pubsys2/xml/en/procedures/99/99-202-008.html)
Procedure Title	Distributors - International
Update Reason	
Date	28-Sep-2006
Section	Section 2 - Maintenance Guidelines
Procedure Number	40-102-004-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-102-004-om-ind.html)
Procedure Title	Tool Requirements
Update Reason	

Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t004-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t004-om.html)
Procedure Title	Air Compressor Air Pressure Rises Slowly
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t005-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t005-om.html)
Procedure Title	Air Compressor Cycles Frequently
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t006-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t006-om.html)
Procedure Title	Air Compressor Noise is Excessive
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t007-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t007-om.html)
Procedure Title	Air Compressor Pumping Excess Lubricating Oil into the Air System
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t008-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t008-om.html)
Procedure Title	Air Compressor Will Not Maintain Adequate Air Pressure (Not Pumping Continuously)
Update Reason	

Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t010-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t010-om.html)
Procedure Title	Air Compressor Will Not Stop Pumping
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t013-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t013-om.html)
Procedure Title	Alternator Not Charging or Insufficient Charging
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t014-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t014-om.html)
Procedure Title	Alternator Overcharging
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t018-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t018-om.html)
Procedure Title	Coolant Contamination
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t020-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t020-om.html)
Procedure Title	Coolant Loss - External
Update Reason	

Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t022-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t022-om.html)
Procedure Title	Coolant Temperature Above Normal - Gradual Overheat
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t023-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t023-om.html)
Procedure Title	Coolant Temperature Above Normal - Sudden Overheat
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t024-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t024-om.html)
Procedure Title	Coolant Temperature Below Normal
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t033-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t033-om.html)
Procedure Title	Engine Acceleration or Response Poor
Update Reason	

Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t043-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t043-om.html)
Procedure Title	Engine Difficult to Start or Will Not Start (Exhaust Smoke)
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t044-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t044-om.html)
Procedure Title	Engine Difficult to Start or Will Not Start (No Exhaust Smoke)
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t047-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t047-om.html)
Procedure Title	Engine Noise Excessive
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t048-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t048-om.html)
Procedure Title	Engine Noise Excessive - Combustion Knocks
Update Reason	
Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t057-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t057-om.html)
Procedure Title	Engine Power Output Low
Update Reason	

Date	29-Sep-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t061-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t061-om.html)
Procedure Title	Engine Runs Rough at Idle
Update Reason	
Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t062-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t062-om.html)
Procedure Title	Engine Runs Rough or Misfires
Update Reason	
Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t064-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t064-om.html)
Procedure Title	Engine Shuts Off Unexpectedly or Dies During Deceleration
Update Reason	
Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t066-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t066-om.html)
Procedure Title	Engine Speed Surges at Low or High Idle
Update Reason	
Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t067-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t067-om.html)
Procedure Title	Engine Speed Surges Under Load or in Operating Range
Update Reason	

Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t072-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t072-om.html)
Procedure Title	Engine Starts But Will Not Keep Running
Update Reason	
Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t075-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t075-om.html)
Procedure Title	Engine Vibration Excessive
Update Reason	
Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t077-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t077-om.html)
Procedure Title	Engine Will Not Crank or Cranks Slowly (Air Starter)
Update Reason	
Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t078-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t078-om.html)
Procedure Title	Engine Will Not Crank or Cranks Slowly (Electric Starter)
Update Reason	
Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t080-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t080-om.html)
Procedure Title	Engine Will Not Reach Rated Speed (RPM)
Update Reason	

Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t081-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t081-om.html)
Procedure Title	Engine Will Not Shut Off
Update Reason	
Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t103-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t103-om.html)
Procedure Title	Lubricating Oil Contaminated
Update Reason	
Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t103-75-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t103-75-om.html)
Procedure Title	Lubricating Oil Loss
Update Reason	
Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t104-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t104-om.html)
Procedure Title	Lubricating Oil Pressure High
Update Reason	
Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t105-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t105-om.html)
Procedure Title	Lubricating Oil Pressure Low
Update Reason	

Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t106-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t106-om.html)
Procedure Title	Lubricating Oil Sludge in the Crankcase Excessive
Update Reason	
Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t116-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t116-om.html)
Procedure Title	Smoke, Black - Excessive
Update Reason	
Date	02-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t122-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t122-om.html)
Procedure Title	Turbocharger Leaks Engine Oil or Fuel
Update Reason	
Date	03-Oct-2006
Section	Front
Procedure Number	4021389-titlepage (/qs3/pubsys2/xml/en/manual/4021389/4021389-titlepage.html)
Procedure Title	Titlepage
Update Reason	
Date	03-Oct-2006
Section	Section D - System Diagrams
Procedure Number	40-200-001-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-200-001-om-ind.html)
Procedure Title	Flow Diagram, Fuel System
Update Reason	

Date	03-Oct-2006
Section	Section D - System Diagrams
Procedure Number	100-200-002 (/qs3/pubsys2/xml/en/procedures/100/100-200-002.html)
Procedure Title	Flow Diagram, Lubricating Oil System
Update Reason	
Date	03-Oct-2006
Section	Section D - System Diagrams
Procedure Number	100-200-003-om-ind (/qs3/pubsys2/xml/en/procedures/100/100-200-003-om-ind.html)
Procedure Title	Flow Diagram, Cooling System
Update Reason	
Date	03-Oct-2006
Section	Section D - System Diagrams
Procedure Number	40-200-004-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-200-004-om-ind.html)
Procedure Title	Flow Diagram, Air Intake System
Update Reason	
Date	03-Oct-2006
Section	Section D - System Diagrams
Procedure Number	40-200-005-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-200-005-om-ind.html)
Procedure Title	Flow Diagram, Exhaust System
Update Reason	
Date	03-Oct-2006
Section	Back
Procedure Number	4021389-backcover (/qs3/pubsys2/xml/en/manual/4021389/4021389-backcover.html)
Procedure Title	Backcover
Update Reason	

Date	09-Oct-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021389-t118-om (/qs3/pubsys2/xml/en/manual/4021389/4021389-t118-om.html)
Procedure Title	Smoke, White - Excessive
Update Reason	
Date	26-Oct-2006
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	100-008-002 (/qs3/pubsys2/xml/en/procedures/100/100-008-002.html)
Procedure Title	Drive Belt, Cooling Fan
Update Reason	
Date	07-Mar-2007
Section	Section E - Engine and System Identification
Procedure Number	40-100-001-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-100-001-om-ind.html)
Procedure Title	Engine Identification
Update Reason	
Date	07-Mar-2007
Section	Section E - Engine and System Identification
Procedure Number	40-100-002-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-100-002-om-ind.html)
Procedure Title	Engine Diagrams
Update Reason	
Date	07-Mar-2007
Section	Section L - Service Literature
Procedure Number	40-205-001-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-205-001-om-ind.html)
Procedure Title	Additional Service Literature
Update Reason	

Date	07-Mar-2007
Section	Section V - Maintenance Specifications
Procedure Number	40-018-015-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-018-015-om-ind.html)
Procedure Title	General Engine
Update Reason	
Date	23-Apr-2007
Section	Section V - Maintenance Specifications
Procedure Number	40-018-016-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-018-016-om-ind.html)
Procedure Title	Fuel System
Update Reason	
Date	23-Apr-2007
Section	Section W - Warranty
Procedure Number	3381322 (/qs3/pubsys2/xml/en/warranty/3381322.html)
Procedure Title	All Engines, International, Industrial (Off-Highway)
Update Reason	
Date	24-Apr-2007
Section	Section V - Maintenance Specifications
Procedure Number	40-018-018-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-018-018-om-ind.html)
Procedure Title	Cooling System
Update Reason	
Date	24-Apr-2007
Section	Section V - Maintenance Specifications
Procedure Number	40-018-019-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-018-019-om-ind.html)
Procedure Title	Air Intake System
Update Reason	

Date	24-Apr-2007
Section	Section V - Maintenance Specifications
Procedure Number	40-018-020-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-018-020-om-ind.html)
Procedure Title	Exhaust System
Update Reason	
Date	25-Apr-2007
Section	Section V - Maintenance Specifications
Procedure Number	40-018-006 (/qs3/pubsys2/xml/en/procedures/40/40-018-006.html)
Procedure Title	Engine Component Torque Values
Update Reason	
Date	21-May-2007
Section	Section V - Maintenance Specifications
Procedure Number	40-018-017-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-018-017-om-ind.html)
Procedure Title	Lubricating Oil System
Update Reason	
Date	20-Jun-2007
Section	Section W - Warranty
Procedure Number	3381321 (/qs3/pubsys2/xml/en/warranty/3381321.html)
Procedure Title	All Engines, United States and Canada, Industrial (Off-Highway)
Update Reason	
Date	02-Jul-2007
Section	Section V - Maintenance Specifications
Procedure Number	40-018-021-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-018-021-om-ind.html)
Procedure Title	Electrical System
Update Reason	

Date	24-Jul-2007
Section	Section i - Introduction
Procedure Number	99-204-008 (/qs3/pubsys2/xml/en/procedures/99/99-204-008.html)
Procedure Title	General Cleaning Instructions
Update Reason	
Date	22-Oct-2007
Section	Section 7 - Maintenance Procedures at 2000 Hours or 2 Years
Procedure Number	99-001-052 (/qs3/pubsys2/xml/en/procedures/99/99-001-052.html)
Procedure Title	Vibration Damper, Viscous
Update Reason	
Date	07-Feb-2008
Section	Section 4 - Maintenance Procedures at 250 Hours or 3 Months
Procedure Number	99-010-028 (/qs3/pubsys2/xml/en/procedures/99/99-010-028.html)
Procedure Title	Charge-Air Piping
Update Reason	
Date	24-Apr-2008
Section	Section S - Service Assistance
Procedure Number	99-202-007 (/qs3/pubsys2/xml/en/procedures/99/99-202-007.html)
Procedure Title	Regional Offices - International
Update Reason	
Date	29-Oct-2008
Section	Section 1 - Operating Instructions
Procedure Number	100-101-004-om-ind (/qs3/pubsys2/xml/en/procedures/100/100-101-004-om-ind.html)
Procedure Title	Cold Weather Starting
Update Reason	

Date	04-Dec-2008
Section	Section S - Service Assistance
Procedure Number	99-202-006 (/qs3/pubsys2/xml/en/procedures/99/99-202-006.html)
Procedure Title	Distributors and Branches
Update Reason	
Date	12-Dec-2008
Section	Section 4 - Maintenance Procedures at 250 Hours or 3 Months
Procedure Number	99-010-059 (/qs3/pubsys2/xml/en/procedures/99/99-010-059.html)
Procedure Title	Air Cleaner Restriction
Update Reason	
Date	16-Jan-2009
Section	Section 5 - Maintenance Procedures at 500 Hours or 6 Months
Procedure Number	40-006-015-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-006-015-om-ind.html)
Procedure Title	Fuel Filter (Spin-On Type)
Update Reason	
Date	25-Feb-2009
Section	Section S - Service Assistance
Procedure Number	99-202-003 (/qs3/pubsys2/xml/en/procedures/99/99-202-003.html)
Procedure Title	Emergency and Technical Service
Update Reason	

Date	14-Apr-2009
Section	Section 6 - Maintenance Procedures at 1000 Hours or 1 Year
Procedure Number	100-008-087-om (/qs3/pubsys2/xml/en/procedures/100/100-008-087-om.html)
Procedure Title	Cooling Fan Belt Tensioner
Update Reason	
Date	14-Apr-2009
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	100-008-087-om-rep (/qs3/pubsys2/xml/en/procedures/100/100-008-087-om-rep.html)
Procedure Title	Cooling Fan Belt Tensioner
Update Reason	
Date	17-Jun-2009
Section	Section 1 - Operating Instructions
Procedure Number	00-101-014-om-ind (/qs3/pubsys2/xml/en/procedures/00/00-101-014-om-ind.html)
Procedure Title	Normal Starting Procedure
Update Reason	
Date	23-Jun-2009
Section	Section 4 - Maintenance Procedures at 250 Hours or 3 Months
Procedure Number	00-012-014-om (/qs3/pubsys2/xml/en/procedures/00/00-012-014-om.html)
Procedure Title	Air Compressor
Update Reason	

Date	25-Jun-2009
Section	Section TS - Troubleshooting Symptoms
Procedure Number	99-t00-001-om (/qs3/pubsys2/xml/en/procedures/99/99-t00-001-om.html)
Procedure Title	Troubleshooting Procedures and Techniques
Update Reason	
Date	22-Jul-2009
Section	Section L - Service Literature
Procedure Number	99-205-004 (/qs3/pubsys2/xml/en/procedures/99/99-205-004.html)
Procedure Title	Cummins Customized Parts Catalog
Update Reason	
Date	10-Aug-2009
Section	Section V - Maintenance Specifications
Procedure Number	100-018-002 (/qs3/pubsys2/xml/en/procedures/100/100-018-002.html)
Procedure Title	Fuel Recommendations and Specifications
Update Reason	
Date	10-Aug-2009
Section	Section V - Maintenance Specifications
Procedure Number	100-018-003 (/qs3/pubsys2/xml/en/procedures/100/100-018-003.html)
Procedure Title	Lubricating Oil Recommendations and Specifications
Update Reason	
Date	10-Sep-2009
Section	Section i - Introduction
Procedure Number	99-204-002-om-ind (/qs3/pubsys2/xml/en/procedures/99/99-204-002-om-ind.html)
Procedure Title	About the Manual
Update Reason	

Date	08-Oct-2009
Section	Section i - Introduction
Procedure Number	99-204-007-om-ind (/qs3/pubsys2/xml/en/procedures/99/99-204-007-om-ind.html)
Procedure Title	General Repair Instructions
Update Reason	
Date	13-Oct-2009
Section	Section 2 - Maintenance Guidelines
Procedure Number	99-102-001 (/qs3/pubsys2/xml/en/procedures/99/99-102-001.html)
Procedure Title	Maintenance Record Form
Update Reason	
Date	13-Oct-2009
Section	Section 4 - Maintenance Procedures at 250 Hours or 3 Months
Procedure Number	99-010-027 (/qs3/pubsys2/xml/en/procedures/99/99-010-027.html)
Procedure Title	Charge-Air Cooler
Update Reason	
Date	14-Oct-2009
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-007-043-om-ind (/qs3/pubsys2/xml/en/procedures/99/99-007-043-om-ind.html)
Procedure Title	Lubricating Oil Level
Update Reason	
Date	15-Oct-2009
Section	Section 2 - Maintenance Guidelines
Procedure Number	40-102-002-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-102-002-om-ind.html)
Procedure Title	Maintenance Schedule
Update Reason	

Date	15-Oct-2009
Section	Section 4 - Maintenance Procedures at 250 Hours or 3 Months
Procedure Number	99-008-045 (/qs3/pubsys2/xml/en/procedures/99/99-008-045.html)
Procedure Title	Radiator Hoses
Update Reason	
Date	17-Nov-2009
Section	Section 2 - Maintenance Guidelines
Procedure Number	99-102-999 (/qs3/pubsys2/xml/en/procedures/99/99-102-999.html)
Procedure Title	Maintenance Guidelines - Overview
Update Reason	
Date	18-Nov-2009
Section	Section 1 - Operating Instructions
Procedure Number	99-101-002 (/qs3/pubsys2/xml/en/procedures/99/99-101-002.html)
Procedure Title	Electromagnetic Interference (EMI)
Update Reason	
Date	18-Nov-2009
Section	Section 7 - Maintenance Procedures at 2000 Hours or 2 Years
Procedure Number	99-001-051 (/qs3/pubsys2/xml/en/procedures/99/99-001-051.html)
Procedure Title	Vibration Damper, Rubber
Update Reason	

Date	19-Nov-2009
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-008-066 (/qs3/pubsys2/xml/en/procedures/99/99-008-066.html)
Procedure Title	Coolant Level
Update Reason	
Date	19-Nov-2009
Section	Section S - Service Assistance
Procedure Number	99-202-002 (/qs3/pubsys2/xml/en/procedures/99/99-202-002.html)
Procedure Title	Routine Service and Parts
Update Reason	
Date	19-Nov-2009
Section	Section S - Service Assistance
Procedure Number	99-202-004 (/qs3/pubsys2/xml/en/procedures/99/99-202-004.html)
Procedure Title	Problem Solving
Update Reason	
Date	06-Jan-2010
Section	Section i - Introduction
Procedure Number	99-204-003-om-ind (/qs3/pubsys2/xml/en/procedures/99/99-204-003-om-ind.html)
Procedure Title	How to Use the Manual
Update Reason	
Date	25-Jan-2010
Section	Section i - Introduction
Procedure Number	99-204-006-om-ind (/qs3/pubsys2/xml/en/procedures/99/99-204-006-om-ind.html)
Procedure Title	General Safety Instructions
Update Reason	

Date	25-Jan-2010
Section	Section 1 - Operating Instructions
Procedure Number	99-101-999-om-ind (/qs3/pubsys2/xml/en/procedures/99/99-101-999-om-ind.html)
Procedure Title	Operating Instructions - Overview
Update Reason	
Date	25-Jan-2010
Section	Section 1 - Operating Instructions
Procedure Number	99-101-008-om-ind (/qs3/pubsys2/xml/en/procedures/99/99-101-008-om-ind.html)
Procedure Title	Engine Operating Range
Update Reason	
Date	25-Jan-2010
Section	Section 1 - Operating Instructions
Procedure Number	99-101-009-om-ind (/qs3/pubsys2/xml/en/procedures/99/99-101-009-om-ind.html)
Procedure Title	Engine Shutdown
Update Reason	
Date	25-Jan-2010
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-103-002 (/qs3/pubsys2/xml/en/procedures/99/99-103-002.html)
Procedure Title	Drive Belts
Update Reason	
Date	25-Jan-2010
Section	Section V - Maintenance Specifications
Procedure Number	99-018-005 (/qs3/pubsys2/xml/en/procedures/99/99-018-005.html)
Procedure Title	Drive Belt Tension
Update Reason	

Date	25-Jan-2010
Section	Section V - Maintenance Specifications
Procedure Number	99-018-009 (/qs3/pubsys2/xml/en/procedures/99/99-018-009.html)
Procedure Title	Capscrew Markings and Torque Values
Update Reason	
Date	04-Feb-2010
Section	Section 1 - Operating Instructions
Procedure Number	99-101-015-om-ind (/qs3/pubsys2/xml/en/procedures/99/99-101-015-om-ind.html)
Procedure Title	Operating the Engine
Update Reason	
Date	19-Mar-2010
Section	Section V - Maintenance Specifications
Procedure Number	00-018-004 (/qs3/pubsys2/xml/en/procedures/00/00-018-004.html)
Procedure Title	Coolant Recommendations and Specifications
Update Reason	
Date	24-Mar-2010
Section	Section i - Introduction
Procedure Number	99-204-009 (/qs3/pubsys2/xml/en/procedures/99/99-204-009.html)
Procedure Title	Acronyms and Abbreviations
Update Reason	
Date	30-Aug-2011
Section	Section V - Maintenance Specifications
Procedure Number	100-018-008 (/qs3/pubsys2/xml/en/procedures/100/100-018-008.html)
Procedure Title	Sealants
Update Reason	Statement added to include sealants not listed in table but specified in a procedure or step.

Date	13-Dec-2011
Section	Section V - Maintenance Specifications
Procedure Number	00-018-004 (/qs3/pubsys2/xml/en/procedures/00/00-018-004.html)
Procedure Title	Coolant Recommendations and Specifications
Update Reason	Updated freezing point chart with the addition of Celsius and Fahrenheit values.
Date	08-Apr-2015
Section	Section 7 - Maintenance Procedures at 2000 Hours or 2 Years
Procedure Number	40-008-018-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-008-018-om-ind.html)
Procedure Title	Cooling System
Update Reason	Added directions to allow the engine to cool before draining coolant.
Date	08-Sep-2016
Section	Section V - Maintenance Specifications
Procedure Number	100-018-024 (/qs3/pubsys2/xml/en/procedures/100/100-018-024.html)
Procedure Title	Cummins®/Fleetguard® Filter Specifications
Update Reason	Updated Cummins® and Fleetguard® part numbers.
Date	12-Dec-2019
Section	Section V - Maintenance Specifications
Procedure Number	40-018-015-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-018-015-om-ind.html)
Procedure Title	General Engine
Update Reason	Added bore specification for the B4.5.

Date	24-Nov-2025
Section	Section 2 - Maintenance Guidelines
Procedure Number	40-102-002-om-ind (/qs3/pubsys2/xml/en/procedures/40/40-102-002-om-ind.html)
Procedure Title	Maintenance Schedule
Update Reason	Added Table 1, Generator and Industry application to procedure.

Last Modified: 08-Dec-2025
